Huawei Access Node YEL-ACC-22 Optical Transport Failure SOP

Node: YEL-ACC-22 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_1535

Description

The Huawei Access Node YEL-ACC-22 is a critical component of the Yelahanka Sub-station, providing high-speed optical transport capabilities to the network. The failure domain of Optical Transport Failure, specifically SFP/XFP degradation, fiber cuts, and Rx/Tx power loss, is a result of the underlying hardware mechanics and protocol operations. The SFP/XFP transceivers used in this equipment are prone to degradation over time due to environmental factors, such as temperature fluctuations and vibration. Furthermore, the fiber connections are susceptible to physical damage, such as fiber cuts, which can disrupt the optical signal. Additionally, the Rx/Tx power loss can be caused by the degradation of the optical components or the presence of signal interference. The Huawei Access Node YEL-ACC-22 uses the OLT (Optical Line Terminal) module, which is responsible for transmitting and receiving optical signals. The OLT module operates at the physical layer, using the ITU-T G.984 standard for GPON (Gigabit Passive Optical Network) transmission. The protocol stack used in this equipment includes the ONU (Optical Network Unit) management protocol, which is responsible for managing the ONU devices connected to the OLT module. In the event of an Optical Transport Failure, the ALARM_1535 alarm is triggered, indicating a warning condition. It is essential to investigate and resolve this issue promptly to prevent further network disruptions.

Pre-requisite Checks

- Verify that the Yelahanka Sub-station is accessible and secure.

- Check the status of the redundant link to ensure that it is operational and can take over in case of a failure.

- Gather all necessary documentation, including the site location, node ID, and vendor information.

- Ensure that all necessary personnel are informed and aware of the situation.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display device, Expected Output: Device Type: Huawei Access Node YEL-ACC-22, Description: This command displays the device type and model number, verifying that the correct equipment is being used.
```

```
Command: display sfp information, Expected Output: SFP Type:
SFP-10G-SR, Description: This command displays the SFP/XFP type and
status, verifying that the correct transceivers are installed and
functioning properly.

Command: display optical-port information, Expected Output: Optical
Port Status: Up, Description: This command displays the optical port
status, verifying that the fiber connections are operational and not
experiencing any issues.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceivers, Command: replace sfp sfp-1 with sfp-10g-sr, Description: This step involves replacing the degraded SFP/XFP transceivers with new ones, ensuring that the optical signal is transmitted correctly.

Step 2: Step: Verify the fiber connections, Command: verify fiber connections, Description: This step involves verifying that the fiber connections are secure and not experiencing any physical damage or signal loss.

Step 3: Step: Adjust the Rx/Tx power, Command: adjust rx-tx power, Description: This step involves adjusting the Rx/Tx power to ensure that the optical signal is transmitted correctly and not experiencing any loss or degradation.

Post-Check Verification

```
Command: display device, Expected Output: Device Type: Huawei Access
Node YEL-ACC-22, Description: This command verifies that the device
type and model number are correct after the resolution steps.

Command: display sfp information, Expected Output: SFP Type:
SFP-10G-SR, Description: This command verifies that the SFP/XFP type
and status are correct after the resolution steps.

Command: display optical-port information, Expected Output: Optical
Port Status: Up, Description: This command verifies that the optical
port status is correct after the resolution steps.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceivers, Command: replace sfp sfp-1 with original-sfp, Description: This step involves restoring the original SFP/XFP transceivers in case the new ones do not resolve the issue.

- Step: Re-adjust the Rx/Tx power, Command: adjust rx-tx power to original setting, Description: This step involves re-adjusting the Rx/Tx power to its original setting in case the new setting does not resolve the issue.